

Euler's Method and Logistic Growth



The Euler's Method formula

- 1 State the recursive formula used in Euler's Method to approximate solutions to $\frac{dy}{dx} = f(x, y)$.
 - 2 Given $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's Method with step size $\Delta x = 0.5$ to estimate $y(1)$.
 - 3 Given $\frac{dy}{dx} = y$ with $y(0) = 1$, use Euler's Method with $\Delta x = 0.5$ to estimate $y(1)$, then compare to the exact solution $y = e^x$.
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Euler's Method with smaller step sizes

- 4 Given $\frac{dy}{dx} = 2x$ with $y(0) = 0$, use Euler's Method with $\Delta x = 0.25$ to estimate $y(0.5)$, then compare to the exact solution $y = x^2$.
 - 5 For $\frac{dy}{dx} = 2x$ with $y(0) = 0$, redo the estimate for $y(0.5)$ using $\Delta x = 0.1$ (5 steps), and compare accuracy to the $\Delta x = 0.25$ result.
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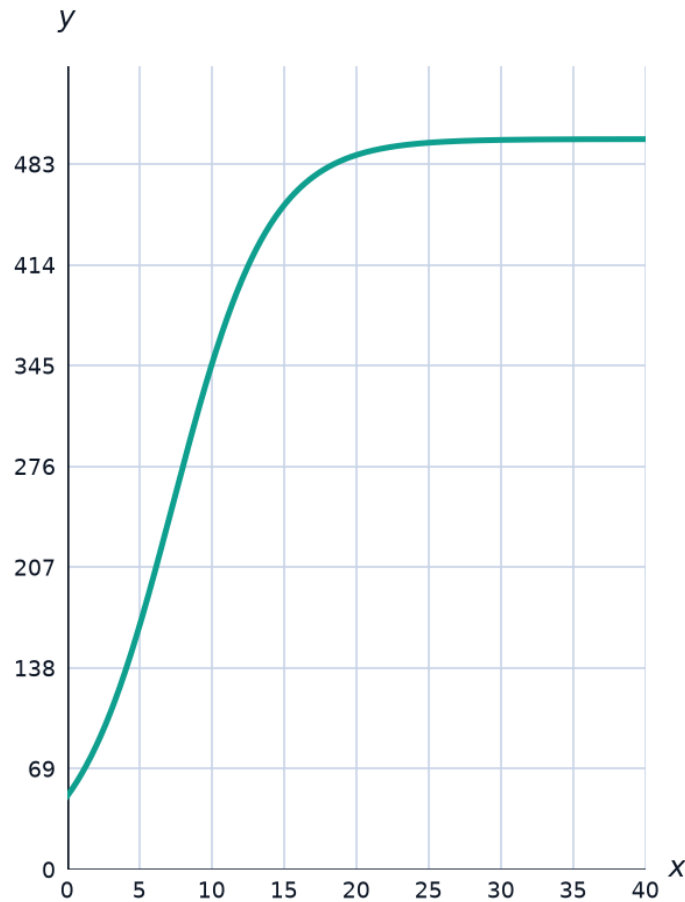
Error analysis in Euler's Method

- 6 Explain why Euler's Method tends to underestimate y when the solution curve is concave up, using a tangent-line reasoning argument.
 - 7 Explain what happens to the accuracy of Euler's Method as the step size $\Delta x \rightarrow 0$.
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Logistic growth models

- 8 State the logistic differential equation and identify the carrying capacity L in $\frac{dP}{dt} = 0.3P(1 - \frac{P}{500})$.
- 9 For the logistic model $\frac{dP}{dt} = 0.3P(1 - \frac{P}{500})$, find the population P at which growth rate is maximized.

- 10 The logistic model with $L = 500$, $k = 0.3$, and $P(0) = 50$ has closed-form solution $P(t) = \frac{500}{1 + 9e^{-0.3t}}$. Sketch its shape and identify the horizontal asymptote.



- 11 For a logistic model with carrying capacity $L = 1000$ and $k = 0.2$, find $\frac{dP}{dt}$ when $P = 200$ and when $P = 800$, and explain which is closer to the maximum growth rate.

Interpreting logistic growth graphically

- 12 Explain what happens to $\frac{dP}{dt}$ as $P \rightarrow L$ in the logistic model, and what this means for the population over time.
- 13 Explain why the logistic model is considered more realistic than the unrestricted exponential model $\frac{dP}{dt} = kP$ for real populations.

Building an Euler's Method table

- 14 Given $\frac{dy}{dx} = x - y$ with $y(0) = 2$, complete an Euler's Method table with $\Delta x = 0.5$ for $x = 0, 0.5, 1, 1.5, 2$, showing each y_n .

- 15 Using the table above, estimate $y(2)$ and state whether the true solution is likely above or below this estimate near $x = 2$, given the slope $\frac{dy}{dx} = x - y$ becomes positive once $y < x$.
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Differential equations as models of real phenomena

- 16 A cup of coffee cools according to Newton's Law of Cooling, $\frac{dT}{dt} = -0.15(T - 20)$, with $T(0) = 90^\circ\text{C}$. Use Euler's Method with $\Delta t = 1$ to estimate $T(2)$.
- 17 Explain why Newton's Law of Cooling, $\frac{dT}{dt} = -k(T - T_{\text{room}})$, has the same mathematical structure as exponential decay shifted by a constant, and state its equilibrium (constant) solution.
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Comparing Euler's Method to the exact logistic solution

- 18 For $\frac{dP}{dt} = 0.3P(1 - \frac{P}{500})$ with $P(0) = 50$, use Euler's Method with $\Delta t = 2$ to estimate $P(2)$, then compare to the exact value $P(2) = \frac{500}{1 + 9e^{-0.6}} \approx 121.9$ (using the closed-form solution from earlier).
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Autonomous differential equations and equilibria

- 19 A differential equation is called autonomous if $\frac{dy}{dx}$ depends only on y , not on x . Explain why both $\frac{dy}{dx} = ky$ and the logistic equation are autonomous, and find the equilibrium solutions of $\frac{dy}{dx} = y(y - 4)$.
- 20 For $\frac{dy}{dx} = y(y - 4)$, determine whether the equilibrium $y = 0$ is stable or unstable by checking the sign of $\frac{dy}{dx}$ just above and below $y = 0$.
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Multiple choice — easy 15

- 21 Given $\frac{dy}{dx} = 2x - 2y$ with $y(2) = -1$, use Euler's Method with step size $\Delta x = 0.25$ to estimate $y(2.25)$.
- a. 2
 - b. -2.5
 - c. 1.5
 - d. 0.5

- 22 Given $\frac{dy}{dx} = 3y$ with $y(2) = 5$, use Euler's Method with step size $\Delta x = 0.5$ to estimate $y(2.5)$.
- a. 12.5
 - b. 20
 - c. -2.5
 - d. 7.5
- 23 Given $\frac{dy}{dx} = x - 3y$ with $y(2) = 0$, use Euler's Method with step size $\Delta x = 0.5$ to estimate $y(2.5)$.
- a. 2
 - b. 3
 - c. 1
 - d. -1
- 24 Given $\frac{dy}{dx} = 2x - 3y$ with $y(1) = -2$, use Euler's Method with step size $\Delta x = 0.1$ to estimate $y(1.1)$.
- a. -0.4
 - b. 0.8
 - c. -2.8
 - d. -1.2
- 25 Given $\frac{dy}{dx} = 2x + 2y$ with $y(1) = 1$, use Euler's Method with step size $\Delta x = 1$ to estimate $y(2)$.
- a. 4
 - b. -3
 - c. 9
 - d. 5

- 26 Given $\frac{dy}{dx} = 3x + 3y$ with $y(2) = 4$, use Euler's Method with step size $\Delta x = 0.25$ to estimate $y(2.25)$.
- a. 4.5
 - b. -0.5
 - c. 13
 - d. 8.5
- 27 Given $\frac{dy}{dx} = x + 3y$ with $y(2) = 4$, use Euler's Method with step size $\Delta x = 0.2$ to estimate $y(2.2)$.
- a. 6.8
 - b. 9.6
 - c. 1.2
 - d. 2.8
- 28 Given $\frac{dy}{dx} = -3x + 3y$ with $y(0) = -2$, use Euler's Method with step size $\Delta x = 0.25$ to estimate $y(0.25)$.
- a. -5
 - b. -3.5
 - c. -0.5
 - d. -1.5
- 29 Given $\frac{dy}{dx} = -2x + y$ with $y(0) = 4$, use Euler's Method with step size $\Delta x = 0.25$ to estimate $y(0.25)$.
- a. 5
 - b. 3
 - c. 1
 - d. 6

- 30 Given $\frac{dy}{dx} = y$ with $y(2) = 1$, use Euler's Method with step size $\Delta x = 0.5$ to estimate $y(2.5)$.
- a. 1.5
 - b. 2
 - c. -0.5
 - d. 0.5
- 31 Given $\frac{dy}{dx} = 3x - y$ with $y(1) = 4$, use Euler's Method with step size $\Delta x = 0.2$ to estimate $y(1.2)$.
- a. 3.8
 - b. -0.2
 - c. 4.2
 - d. 3.6
- 32 Given $\frac{dy}{dx} = -3x + 3y$ with $y(1) = 0$, use Euler's Method with step size $\Delta x = 0.1$ to estimate $y(1.1)$.
- a. 0.3
 - b. -0.6
 - c. -0.3
 - d. 1.7
- 33 Given $\frac{dy}{dx} = -2x + 3y$ with $y(0) = 5$, use Euler's Method with step size $\Delta x = 0.2$ to estimate $y(0.2)$.
- a. 8
 - b. 3
 - c. 11
 - d. 2

- 34 Given $\frac{dy}{dx} = -x + y$ with $y(1) = -3$, use Euler's Method with step size $\Delta x = 0.2$ to estimate $y(1.2)$.
- a. -4.6
 - b. -2.2
 - c. -3.8
 - d. -0.8
- 35 Given $\frac{dy}{dx} = 3x + 2y$ with $y(0) = -2$, use Euler's Method with step size $\Delta x = 1$ to estimate $y(1)$.
- a. -4
 - b. -6
 - c. -10
 - d. 2
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Multiple choice — medium 15

- 36 Given $\frac{dy}{dx} = -2x - y$ with $y(0) = 4$, use Euler's Method with step size $\Delta x = 0.5$ to estimate $y(1)$ (two steps).
- a. 0.5
 - b. 1
 - c. 2
 - d. -0.5
- 37 Given $\frac{dy}{dx} = -2x - 2y$ with $y(0) = -4$, use Euler's Method with step size $\Delta x = 0.5$ to estimate $y(1)$ (two steps).
- a. 0
 - b. -1.5
 - c. -0.5
 - d. 0.5

- 38 Given $\frac{dy}{dx} = 2x - 2y$ with $y(0) = 0$, use Euler's Method with step size $\Delta x = 1$ to estimate $y(2)$ (two steps).
- a. -2
 - b. 0
 - c. 2
 - d. 3
- 39 Given $\frac{dy}{dx} = -2x + y$ with $y(0) = -4$, use Euler's Method with step size $\Delta x = 0.5$ to estimate $y(1)$ (two steps).
- a. -6
 - b. 9.5
 - c. -9
 - d. -9.5
- 40 Given $\frac{dy}{dx} = -x + y$ with $y(0) = 3$, use Euler's Method with step size $\Delta x = 0.5$ to estimate $y(1)$ (two steps).
- a. 4.5
 - b. 7
 - c. -6.5
 - d. 6.5
- 41 Given $\frac{dy}{dx} = -x - 2y$ with $y(0) = 3$, use Euler's Method with step size $\Delta x = 1$ to estimate $y(2)$ (two steps).
- a. 2
 - b. -2
 - c. 3
 - d. -3

- 42 Given $\frac{dy}{dx} = 2x$ with $y(0) = 1$, use Euler's Method with step size $\Delta x = 1$ to estimate $y(2)$ (two steps).
- a. 4
 - b. 3
 - c. -3
 - d. 1
- 43 Given $\frac{dy}{dx} = -2x + 2y$ with $y(0) = 1$, use Euler's Method with step size $\Delta x = 0.5$ to estimate $y(1)$ (two steps).
- a. -3.5
 - b. 2
 - c. 4
 - d. 3.5
- 44 Given $\frac{dy}{dx} = 2x - y$ with $y(0) = 3$, use Euler's Method with step size $\Delta x = 1$ to estimate $y(2)$ (two steps).
- a. 3
 - b. 2
 - c. 0
 - d. -2
- 45 Given $\frac{dy}{dx} = -2y$ with $y(0) = -1$, use Euler's Method with step size $\Delta x = 0.5$ to estimate $y(1)$ (two steps).
- a. 0
 - b. -1
 - c. -0.5
 - d. 0.5

- 46 Given $\frac{dy}{dx} = 2x + y$ with $y(0) = -1$, use Euler's Method with step size $\Delta x = 0.25$ to estimate $y(0.5)$ (two steps).
- a. -1.4375
 - b. -1.1875
 - c. -1.25
 - d. 1.4375
- 47 Given $\frac{dy}{dx} = x$ with $y(0) = 0$, use Euler's Method with step size $\Delta x = 0.5$ to estimate $y(1)$ (two steps).
- a. 0.75
 - b. 0.25
 - c. -0.25
 - d. 0
- 48 Given $\frac{dy}{dx} = y$ with $y(0) = -1$, use Euler's Method with step size $\Delta x = 0.25$ to estimate $y(0.5)$ (two steps).
- a. -1.5625
 - b. -1.3125
 - c. -1.25
 - d. 1.5625
- 49 Given $\frac{dy}{dx} = -2x - y$ with $y(0) = 0$, use Euler's Method with step size $\Delta x = 0.25$ to estimate $y(0.5)$ (two steps).
- a. -0.125
 - b. -1.125
 - c. 0
 - d. 0.125

- 50 Given $\frac{dy}{dx} = -2x - y$ with $y(0) = -3$, use Euler's Method with step size $\Delta x = 0.25$ to estimate $y(0.5)$ (two steps).
- a. -2.25
 - b. 1.8125
 - c. -1.5625
 - d. -1.8125
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Multiple choice — hard 15

- 51 For the logistic model $\frac{dP}{dt} = \frac{1}{10}P\left(1 - \frac{P}{500}\right)$, find $\frac{dP}{dt}$ when $P = 350$.
- a. $\frac{21}{2}$
 - b. $\frac{11}{2}$
 - c. 35
 - d. $\frac{31}{2}$
- 52 For the logistic model $\frac{dP}{dt} = \frac{1}{4}P\left(1 - \frac{P}{500}\right)$, find $\frac{dP}{dt}$ when $P = 250$.
- a. $\frac{105}{4}$
 - b. $\frac{145}{4}$
 - c. $\frac{125}{2}$
 - d. $\frac{125}{4}$
- 53 For the logistic model $\frac{dP}{dt} = \frac{1}{10}P\left(1 - \frac{P}{400}\right)$, find $\frac{dP}{dt}$ when $P = 50$.
- a. 5
 - b. $\frac{35}{8}$
 - c. $\frac{75}{8}$
 - d. $-\frac{5}{8}$

- 54 For the logistic model $\frac{dP}{dt} = \frac{3}{10}P\left(1 - \frac{P}{400}\right)$, find $\frac{dP}{dt}$ when $P = 150$.
- a. $\frac{265}{8}$
 - b. 45
 - c. $\frac{185}{8}$
 - d. $\frac{225}{8}$
- 55 For the logistic model $\frac{dP}{dt} = \frac{1}{10}P\left(1 - \frac{P}{100}\right)$, find $\frac{dP}{dt}$ when $P = 50$.
- a. $\frac{5}{2}$
 - b. $-\frac{5}{2}$
 - c. 5
 - d. $\frac{15}{2}$
- 56 A cup of coffee cools according to $\frac{dT}{dt} = -0.1(T - 20)$, with $T(0) = 88^\circ\text{C}$. Use Euler's Method with $\Delta t = 1$ to estimate $T(2)$.
- a. 81.2
 - b. 80.08
 - c. 70.08
 - d. 75.08
- 57 A cup of coffee cools according to $\frac{dT}{dt} = -0.15(T - 25)$, with $T(0) = 79^\circ\text{C}$. Use Euler's Method with $\Delta t = 0.5$ to estimate $T(1)$.
- a. 76.2
 - b. 74.95
 - c. 71.2
 - d. 66.2

- 58 A cup of coffee cools according to $\frac{dT}{dt} = -0.1(T - 20)$, with $T(0) = 77^\circ\text{C}$. Use Euler's Method with $\Delta t = 1$ to estimate $T(2)$.
- a. 66.17
 - b. 71.3
 - c. 61.17
 - d. 71.17
- 59 A cup of coffee cools according to $\frac{dT}{dt} = -0.2(T - 21)$, with $T(0) = 87^\circ\text{C}$. Use Euler's Method with $\Delta t = 1$ to estimate $T(2)$.
- a. 63.24
 - b. 73.8
 - c. 68.24
 - d. 58.24
- 60 A cup of coffee cools according to $\frac{dT}{dt} = -0.3(T - 20)$, with $T(0) = 97^\circ\text{C}$. Use Euler's Method with $\Delta t = 1$ to estimate $T(2)$.
- a. 57.73
 - b. 73.9
 - c. 62.73
 - d. 52.73
- 61 For $\frac{dy}{dx} = y(y + 8)$, determine whether the equilibrium $y = 0$ is stable or unstable.
- a. Cannot be determined without solving the equation exactly
 - b. Unstable
 - c. Stable
 - d. Semi-stable (attracting from one side only)

- 62 For $\frac{dy}{dx} = y(y - 7)$, determine whether the equilibrium $y = 0$ is stable or unstable.
- a. Stable
 - b. Semi-stable (attracting from one side only)
 - c. Cannot be determined without solving the equation exactly
 - d. Unstable
- 63 For $\frac{dy}{dx} = y(y - 6)$, determine whether the equilibrium $y = 0$ is stable or unstable.
- a. Cannot be determined without solving the equation exactly
 - b. Semi-stable (attracting from one side only)
 - c. Stable
 - d. Unstable
- 64 For $\frac{dy}{dx} = y(y + 3)$, determine whether the equilibrium $y = 0$ is stable or unstable.
- a. Unstable
 - b. Cannot be determined without solving the equation exactly
 - c. Semi-stable (attracting from one side only)
 - d. Stable
- 65 For $\frac{dy}{dx} = y(y + 7)$, determine whether the equilibrium $y = 0$ is stable or unstable.
- a. Semi-stable (attracting from one side only)
 - b. Unstable
 - c. Cannot be determined without solving the equation exactly
 - d. Stable